Airport Data Integration - Team 5

Zheng Fu, Jing Cheng, Bo-Ye Li, Chun Huang-Lin, and Niklas Dziwisch

University of Mannheim
Web Data Integration - HWS 2026

1 Description of the Use Case

The goal of this project is to integrate heterogeneous airport data from three independent Web sources into a single, unified dataset. The selected sources are OurAirports, Wikidata, queried via the QLever SPARQL endpoint, and a global airport dataset from Kaggle. All three sources describe the same real-world entity type, namely airports, but differ in their coverage, schema, attribute naming, and level of detail. The integrated dataset should provide a consistent representation of airports by combining complementary information such as airport names, IATA and ICAO codes, geographic coordinates, country and city information, time zones, elevation, and runway information. Runways from OurAirports are represented as a list attribute. By integrating these sources, the project aims to resolve corresponding airport records across datasets and to combine their information into a more complete and consistent representation of each airport.

2 Fulfillment of Project Requirements

2.1 Dataset Overview

Table 1 summarizes the three input datasets, sources, format, number of entities, and attributes.

Table 1: Datasets

Dataset	**Source (*)**	**Format**	**# of entities**	**# of attr.**	**List of attributes (**)**
Global Airports Dataset	Kaggle	csv	6,391	13	AirportName, IATA, ICAO, TimeZone, City_Name, City_IATA, UTC_Offset_Hours, UTC_Offset_Seconds, Country_CodeA2, Country_CodeA3, Country_Name, GeoPointLat, GeoPointLong
Wikidata	SPARQL via QLever (Uni Freiburg)	csv	24,871	13	airport, name, country, elevation, faa (MV), iata (MV), icao (MV), located_in, operator (MV), owner (MV), timezone (MV), lat, lon
OurAirports	ourairports.com	csv	86,151	19	id, ident, type, name, latitude_deg, longitude_deg, elevation_ft, continent (MV), iso_country, iso_region, municipality, scheduled_service, icao_code (MV), iata_code (MV), gps_code (MV), local_code (MV), home_link (MV), wikipedia_link (MV), runways (MV)

(*) Kaggle: static CSV download. Wikidata: SPARQL query via the QLever endpoint. OurAirports: static CSV bulk download, augmented with a joined runway list from `runways.csv`.

(**) Attributes with more than 30% missing values are marked with "(MV)".

2.2 Data Profiling of Each Dataset

Global Airports Dataset (Kaggle) The first data source of airport data is Global Airports Dataset from Kaggle website. This dataset essentially contains basic airport profile data such as name and geometry, and identity code assigned by international aviation institutes. The schema of this dataset is shown in Table 2. Reasonably, the ICAO code should uniquely identify one airport and therefore can serve as the gold-standard set for evaluation. However, two ICAO codes were found to be duplicated across four records. We therefore kept only one record for each code, and the number of entities has been adjusted to 6,391.

Table 2: Data profile of the Global Airports Dataset (Kaggle)

Attribute	Type	Description	Missing values
ICAO	string	4-letter ICAO code	20.37%
AirportName	string	Airport name	0%
GeoPointLat	float	Latitude (decimal deg.)	0%
GeoPointLong	float	Longitude (decimal deg.)	0%
UTC_Offset_Hours	float	UTC offset, hours	1.38%
UTC_Offset_Seconds	float	UTC offset, seconds	1.38%
IATA	string	3-letter IATA code	0.31%
TimeZone	string	IANA timezone ID	1.38%
City_Name	string	City name	0%
City_IATA	string	City IATA code	0.97%
Country_CodeA2	string	2-letter country code	0.22%
Country_CodeA3	string	3-letter country code	0%
Country_Name	string	Country name	0%

Wikidata A further source is Wikidata, queried through the QLever SPARQL engine (University of Freiburg) due to throughput limits of the official endpoint. Because Wikidata is collaboratively edited, attributes often carry conflicting or redundant statements. Attribute conflicts were resolved per entity: genuinely multi-valued attributes (operator, owner) were kept as lists, others resolved to their mode, and coordinates resolved as a single (latitude, longitude) pair to avoid combining mismatched axes. Table 3 reports, per attribute, the missing rate and conflict rate (proportion of entities with multiple distinct raw values before resolution). Apart from these attribute conflicts, we also observed duplicate identifiers across records: records sharing an IATA/ICAO code were merged if within 1 km of each other; otherwise, the shared code was set to missing.

Table 3: Data profile of the Wikidata airport dataset

Attribute	Type	Description	Missing values	Conflict rate
airport	string	Wikidata entity URI; serves as the primary key	0%	–
name	string	English name of the airport	0%	–
country	string	Country in which the airport is located	0.06%	0.21%
located_in	string	Administrative territorial entity the airport is located in	1.48%	3.81%
lat, lon	float	Coordinates (decimal degrees), resolved as a pair	0.21%	21.21%
elevation	float	Elevation above sea level, normalized to metres	24.47%	2.09%

Attribute	Type	Description	Missing values	Conflict rate
icao	string	4-letter ICAO airport code	**53.52**%	0.04%
faa	string	FAA airport code (US-specific identifier)	**55.60**%	0.00%
timezone	string	Time zone of the airport's location	**59.03**%	28.08%
iata	string	3-letter IATA airport code	**67.96**%	0.01%
operator	list	Operating organization(s); list-valued	**88.46**%	–
owner	list	Owning organization(s); list-valued	**98.50**%	–

OurAirports Our third source of airport data is OurAirports, a community-maintained open database whose content derives largely from official aviation publications (FAA, DAFIF) supplemented by user contributions. We used two files: the airport table (86,151 records, 19 attributes) and the runway table (48,278 records, 20 attributes). The source is dominated by minor facilities: small airports (49.6%) and heliports (27.0%) make up over three quarters of all records, whereas medium and large airports account for only 4.8% and 1.4%, respectively. A further 13,548 records (15.7%) are marked as closed. These were not removed at this stage, as whether to exclude them is deferred to the identity resolution and data fusion steps.

Table 4: Data profile of the OurAirports airport dataset

Attribute	Type	Description	Missing values	Distinct values
id	int	Internal OurAirports numeric identifier	0%	100%
ident	string	OurAirports airport identifier; serves as the primary key	0%	100%
type	string	Facility category (e.g. small, medium or large airport, heliport, closed)	0%	0.01%
name	string	Name of the airport	0%	94.67%
latitude_deg, longitude_deg	float	Coordinates (decimal degrees)	0%	99.84%
continent	string	2-letter continent code	46.22%	0.01%
iso_country	string	ISO 3166-1 alpha-2 country code	0.35%	0.29%
iso_region	string	ISO 3166-2 region code	0%	3.50%
scheduled_service	string	Whether the airport has scheduled airline service (yes/no)	0%	<0.01%
municipality	string	Primary municipality served by the airport	5.51%	47.18%
elevation_ft	float	Elevation above sea level, in feet	17.35%	9.10%
gps_code	string	Code used in GPS databases	48.42%	100%
runways	list	Runways of the airport (length, width, surface, lighting, closure status, end identifiers); list-valued	**52.22**%	–
local_code	string	National or local airport code (e.g. FAA identifier)	**58.09**%	95.84%
wikipedia_link	string	URL of the airport's Wikipedia article	**80.55**%	99.27%
icao_code	string	4-letter ICAO airport code	**87.78**%	100%
iata_code	string	3-letter IATA airport code	**89.49**%	100%
home_link	string	URL of the airport's official website	**94.47**%	98.93%

2.3 Integrated Schema and Overlap with Input Schemata

Integrated Schema The integrated schema contains 12 conceptual attributes and therefore satisfies Requirement 4, which requires at least eight attributes. It also satisfies Requirement 5: nine attributes occur in at least two datasets, and five non-name attributes including iata, icao, coordinates, country, and city/municipality occur in all three

datasets. The runway information additionally provides a list-valued attribute by joining the one-to-many records in `runways.csv` to OurAirports entities via `airport_ident`. Table 5 elaborates the objective integrated schema and attribute intersection analysis. The full schema definition in `xsd` format is available here.

Table 5: Attribute Intersection with Integrated Schema

Attribute name	**Attribute type**	**Datasets in which the attribute is found**
name	string	Kaggle, Wikidata, OurAirports
iata	string	Kaggle, Wikidata, OurAirports
icao	string	Kaggle, Wikidata, OurAirports
coordinates	pair of floats	Kaggle, Wikidata, OurAirports
country	string	Kaggle, Wikidata, OurAirports
city/municipality	string	Kaggle, Wikidata, OurAirports
elevation	float	Wikidata, OurAirports
timezone	string	Kaggle, Wikidata
local/FAA code	string	Wikidata, OurAirports
operator	list	Wikidata
owner	list	Wikidata
runways	list	OurAirports

Entity Overlap To evaluate whether overlapping entities are sufficient, we estimated overlap using ICAO code as a nominally unique identifier, computing pairwise and three-way intersections as well as the union across all three sources. ICAO was chosen over IATA and FAA for this estimate because it has the broadest coverage across both commercial and general-aviation airports; all three codes are excluded from the identity resolution step itself and instead serve as a gold-standard set for evaluating the similarity-based matcher. Table 6 reports these counts as a lower-bound estimate: the true overlap is expected to be higher once name/coordinate matching also catches entities with missing or inconsistent codes.

Table 6: Entity Overlap Across Datasets (based on ICAO code)

		Overlap ($\cap$) with		
Dataset	**Distinct ICAO codes**	**OurAirports**	**Kaggle**	**Wikidata**
OurAirports	10,530	–	4,700	7,758
Kaggle	5,089	4,700	–	4,524
Wikidata	11,490	7,758	4,524	–

Entities present in all three datasets: 4,350. Union across all three: 14,477 (Requirement 2 target: $\geq$2,500). Entities present in at least two datasets: 8,282 (Requirement 3 target: $\geq$1,000).

Since ICAO/IATA/FAA codes are reserved for evaluation only, identity resolution relies on a combination of name, coordinates, and location attributes instead. Possible combinations of identifying attributes include (name, coordinates) and (name, city/municipality, country), potentially combined in a tiered strategy where the latter serves as a fallback when coordinates are missing or unreliable.